

EDIBLE ELECTRONICS

A Joke Or A Cure?



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Implantable and ingestible (edible) electronic devices that stay inside your body pose unique engineering challenges. These should not only use the best processor but also be minimal in size, biocompatible, safe and extremely reliable. Reliability is all the more important as it often turns out to be a case of life or death!

Despite so many risks and challenges, it is awesome to see how implants have developed since the days of the first pacemaker in 1958. From cardiac pacemakers to cochlear implants, from brain interfaces to retinal implants, there are numerous implantable medical electronic devices available today. Even more exciting is the emerging field of edibles—tiny, capsule-sized electronic devices that are consumed orally for diagnosis and treatment of diseases. Some edibles are designed to remain inside the body for some time, while others do their job and get disposed of within minutes.

Today, we have reached a state where inventions like these no longer surprise us

because our minds have become tuned to a sci-fi future, and we have started expecting such developments. So let us put aside the wow-factor, and instead look at the current and future state of implants and edibles.

Painless diabetes testing, drug delivery and more

With improvements in quality and reliability, there is now a reasonably good demand for cochlear, retinal and cardiovascular implants. Cardiovascular implants have evolved much in recent years, and have overcome past constraints regarding compatibility with imaging systems such as magnetic resonance imaging (MRI). Interestingly, the rise in lifestyle diseases like diabetes has also led to an increased demand for implantable devices like implantable continuous glucose monitoring and implantable infusion pumps. There are also implantable devices for phrenic nerve stimulation to restore breathing function in patients, and sacral nerve stimulation for patients with bladder disorders. Implantable neuro-stimulators, on the other hand, help those with neurological disorders like Parkinson's disease.

With the availability of better biocompatible materials that minimise the possibility of infections, there is greater faith in implants. Researchers at the Graeme Clark Institute have developed an implant that is fitted under the scalp to diagnose and treat epilepsy, and even forecast a likely seizure. They are also developing an implantable drug pump for patients with drug-resistant epilepsy.

Cochlear implants are used when hearing aids don't work well—that is, when the patient has severe sensorineural hearing loss due to absent or reduced cochlear hair cell function. The implant basically carries out the function of the cochlea or inner ear, stimulating the auditory nerve directly. Retinal implants are giving vision to

The journey of an ingestible sensor
(Credit: Proteus Digital Health)

